

RIGOR-OMR: Relation-Aware Interpretable Geometry for Zero-Shot Cross-Domain Optical Music Recognition

Anonymous submission

Abstract

Optical music recognition (OMR) systems achieve strong performance on in-domain benchmarks, yet whether such results translate into plug-and-play recognition on unseen score domains remains underexplored. We present RIGOR-OMR (Relation-aware Interpretable Geometry for Out-of-domain Recognition), a modular framework that leverages staff-normalized geometry to guide visual symbol detection and explicit relation reasoning, while reconstructing structured outputs without target-domain training or dataset-specific adaptation. To compare systems with heterogeneous outputs, we introduce a representation-agnostic pitch–duration event evaluation protocol. On Polish Scores, RIGOR-OMR achieves 17.083 Event-F1, representing relative improvements of 357.8% over SMT and 1182.4% over Zeus. On Debussy, RIGOR-OMR significantly outperforms SMT while performing comparably to Zeus. Ablation studies confirm that explicit relation reasoning is the primary source of these improvements. Extensive experiments across Polish Scores, Debussy, GrandStaff, and OLiMPiC establish RIGOR-OMR as an effective approach to zero-shot music-content recognition under substantial domain shift.

1 Introduction

Optical music recognition converts score images into machine-readable notation. Unlike ordinary character recognition, the meaning of a mark depends jointly on its class, staff-relative position, attachment relations, voice, and reading order (Calvo-Zaragoza, Jr., and Pacha 2021; Rebelo et al. 2012). Modern systems increasingly treat OMR as image-to-sequence prediction, using Transformer decoders to emit compact musical encodings (Vaswani et al. 2017; Ríos-Vila, Calvo-Zaragoza, and Paquet 2024; Li et al. 2023). This formulation has enabled recognition beyond monophonic incipits and toward piano systems or complete pages (Ríos-Vila, Calvo-Zaragoza, and Rizo 2023; Ríos-Vila et al. 2026). Yet the apparent progress hides a deployment problem: a model trained on one renderer, score collection, vocabulary, or page layout can fail sharply on another.

The shift is visible before any model is run. Synthetic piano pages have clean backgrounds, regular spacing, and con-

strained engraving, whereas scanned historical scores contain uneven contrast, page texture, warped systems, typography changes, and more varied annotations. The output space shifts as well: Humdrum-based encodings, Linearized MusicXML (LMX), tuplets, voices, and staff markers differ in vocabulary and structural assumptions. General transfer-learning results predict that higher model layers become increasingly task-specific (Pan and Yang 2010; Yosinski et al. 2014); for OMR, the decoder and notation vocabulary are precisely the most domain-specific components.

This paper introduces RIGOR-OMR, short for *Relation-aware Interpretable Geometry for Out-of-domain Recognition*. Rather than relying exclusively on a source-specific autoregressive vocabulary, RIGOR-OMR exposes a modular recognition path: staff geometry establishes a normalized coordinate frame; DEIM-D-FINE detects symbols; optional SAM2 masks refine local shape evidence; explicit relations connect noteheads, stems, and beams; and a deterministic backend exports MusicXML or an evaluation-compatible sequence. The intermediate artifacts make it possible to distinguish detection, pitch, duration, relation, and serialization failures.

Our contributions are fourfold:

- We formulate practical OMR applicability as zero-shot transfer under matched inputs, frozen target-domain decisions, and a common content projection, rather than extrapolating from in-domain accuracy.
- We introduce RIGOR-OMR, an inspectable geometry–detection–relation–reconstruction pipeline whose detector can be replaced without changing the structured backend.
- Across Polish and Debussy, local runs against SMT and Zeus show that RIGOR-OMR is significantly stronger than both baselines on Polish and significantly stronger than SMT while statistically comparable to Zeus on Debussy.
- Full-factorial and edge-level ablations isolate explicit relation reasoning as the main transferable mechanism, while strict OLiMPiC metrics identify global ordering and hierarchy as the remaining limitation.

2 Related Work

2.1 Datasets and Modular OMR

OMR research has historically decomposed a page into staff detection, symbol recognition, and notation reconstruction

(Calvo-Zaragoza, Jr., and Pacha 2021; Rebelo et al. 2012). Dataset design reflects this decomposition. MUSCIMA++ provides handwritten symbols and explicit pairwise relations, enabling notation-graph experiments (Jr. and Pecina 2017). DeepScores and DeepScoresV2 provide dense synthetic annotations for tiny printed objects, instance segmentation, and detection (Tuggenier et al. 2018, 2021). Camera-PrIMuS supports end-to-end recognition under realistic image degradation but is primarily monophonic (Calvo-Zaragoza and Rizo 2018). RealScores and recent instance-segmentation studies further emphasize that real-world score images differ materially from synthetic renders (Shatri and Fazekas 2023, 2024).

Object-centric OMR benefits from detectors that preserve small, dense instances. DETR introduced set prediction with bipartite matching (Carion et al. 2020); RT-DETRv2, D-FINE, and DEIM improve multiscale sampling, localization, and convergence (Lv et al. 2024; Peng et al. 2025; Huang et al. 2025). Promptable segmentation offers a complementary route from boxes to boundaries. SAM established zero-shot promptable segmentation (Kirillov et al. 2023), while SAM2 extended the design and improved image efficiency (Ravi et al. 2025). RIGOR-OMR uses these models as reusable components, not as evidence that natural-image priors alone understand notation semantics. DINOv2 features support retrieval and crop-level auxiliary classification (Oquab et al. 2024), but all final claims are based on end-output metrics rather than embedding similarity.

2.2 End-to-End Pianoform Recognition

GrandStaff introduced a large pianoform corpus and a sequence-recognition formulation over kern-like encodings (Ríos-Vila, Calvo-Zaragoza, and Rizo 2023). Sheet Music Transformer extended image-to-sequence OMR beyond monophonic transcription (Ríos-Vila, Calvo-Zaragoza, and Paquet 2024), and full-page variants combined convolutional encoders, autoregressive decoders, synthetic curriculum learning, and target-domain fine-tuning (Ríos-Vila et al. 2026). TrOMR likewise targets polyphonic recognition with Transformer-based decoding (Li et al. 2023). OLIMPiC introduced LMX, aligned MusicXML-compatible output, scanned evaluation systems, and a tree-aware metric (Mayer et al. 2024). These systems demonstrate strong in-domain accuracy, but their results also motivate a controlled view of cross-domain behavior.

2.3 Transfer and Domain Shift

Transfer learning studies distinguish reusable lower-level representations from source-specific higher layers (Pan and Yang 2010; Yosinski et al. 2014). Domain-adversarial training seeks features that cannot discriminate source from target domains (Ganin et al. 2016), and domain-generalization work organizes interventions into data, representation, and learning strategies (Wang et al. 2021). RIGOR-OMR studies a narrower and practically common setting: an OMR system must operate on an unseen collection without target annotations, while appearance, layout, and output representation may shift simultaneously.

3 Methodology

Figure 1 summarizes the pipeline. Each stage writes inspectable artifacts, allowing an end-output error to be traced to geometry, detection, relation assignment, or structured export.

3.1 Staff Geometry Estimation

Given a score image I , horizontal projections and connected components identify five-line staff groups. For staff k , the line coordinates $\{\ell_{k,j}\}_{j=1}^5$ define the robust spacing

$$s_k = \text{median}_{j=1}^4 (\ell_{k,j+1} - \ell_{k,j}). \quad (1)$$

The lines and inter-line half spaces form a discrete pitch grid G_k . We express box sizes, attachment distances, and prompt offsets in units of s_k , so the downstream rules remain comparable across resolution and engraving scale. The geometry stage outputs staff regions, line coordinates, normalized symbol coordinates, and candidate pitch-grid positions.

3.2 Staff-Aware Symbol Detection

The score image is processed by a DEIM-D-FINE detector. Its HGNetv2 backbone and hybrid encoder produce multiscale features, while symbol queries predict class-labeled boxes (Huang et al. 2025; Peng et al. 2025). Full-page and staff-crop predictions are merged by class-aware non-maximum suppression. Each detection $d_i = (b_i, c_i, q_i)$ contains a box b_i , symbol class c_i , and confidence q_i ; overlap with the estimated staff regions assigns it to a staff and maps its center to normalized grid coordinates. A lightweight crop classifier may relabel only within a compatible family, preventing a clef classifier, for example, from changing a notehead into a rest.

3.3 Staff-Aware Prompt Generation and Mask Refinement

For detection d_i , the prompt generator constructs

$$\mathcal{P}_i = (b_i, p_i^+, \mathcal{P}_i^-), \quad (2)$$

where b_i is the box prompt, p_i^+ is a foreground point, and $\mathcal{P}_i^-$ contains staff-conditioned negative points. Prompt placement depends on the predicted family: noteheads tolerate staff-line overlap, whereas elongated stems and beams use orientation-aware support. The image crop and prompt set are passed to the SAM2 image, prompt, and mask encoders (Ravi et al. 2025). SAM2 is optional and acts as a boundary refiner rather than a notation classifier; this distinction matches the factorial ablation in Section 4.5, where masks improve inspectability but not the final Event-F1.

3.4 Geometric Primitive Extraction and Relation Parsing

Each refined instance mask is converted into an appropriate primitive: polygons for area-like noteheads and accidentals, and pruned skeletons or endpoints for stems, beams, ledger lines, barlines, and slurs. Every primitive becomes a typed node $v \in V$ with class, staff index, box, mask evidence, and staff-normalized coordinates. RIGOR-OMR then

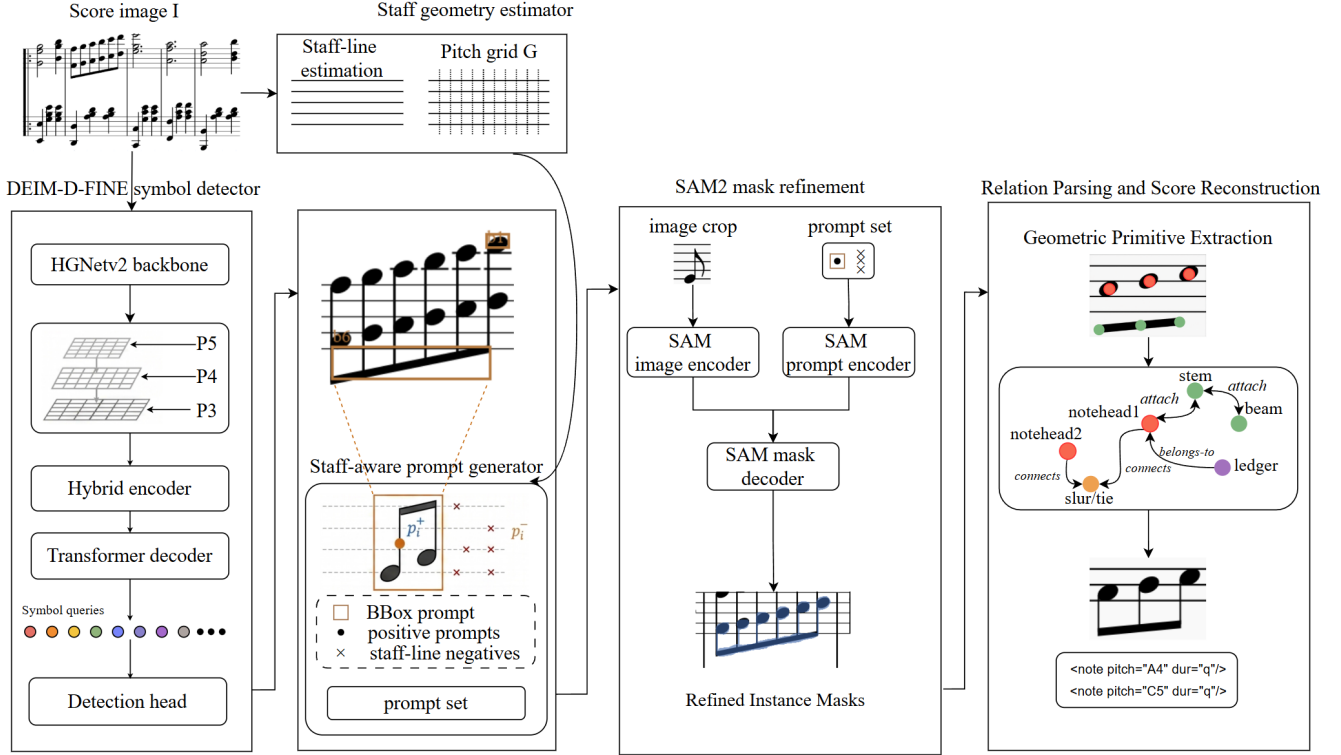


Figure 1: Overview of RIGOR-OMR. A staff-geometry estimator defines a normalized pitch grid; DEIM-D-FINE detects notation symbols; staff-aware prompts guide optional SAM2 mask refinement; and typed geometric relations support pitch–duration event assembly and structured score reconstruction.

constructs $G = (V, E)$ using class-compatible geometric candidates. Notehead–stem edges require horizontal compatibility and a bounded endpoint distance; stem–beam edges require proximity between a stem endpoint and beam support. Ledger–notehead, accidental–notehead, and slur/tie endpoint candidates use analogous type-specific constraints. Conflict pruning retains a consistent set of attachments before event assembly.

3.5 Global Structure Decoder

The upgraded reconstruction backend converts connected primitives into musical events and then decodes a canonical score order; it is a deterministic backend upgrade rather than an additional learned model. The clef-conditioned grid G_k determines pitch, while stem, beam, flag, and dot evidence determines duration, and relation edges bind these attributes to the same event. The decoder groups staves into piano systems, obtains shared measure partitions from supported barlines, clusters staff-normalized horizontal coordinates into onset hypotheses, groups compatible simultaneous notes into chords, and serializes in system–measure–onset–staff order with deterministic pitch and geometry tie breaking. It exports both an auditable event representation and MusicXML or a dataset-compatible proxy sequence. Because decoding is deterministic once V and E are fixed, relation families can be removed and the score rebuilt without retraining or

rerunning the detector, enabling the controlled edge ablations in Section 4.6.

4 Experiments

4.1 Problem Formulation and Evaluation Protocol

Let $\mathcal{D}_s$ denote the source domain used to train a recognition component and $\mathcal{D}_t = \{(x_i^t, y_i^t)\}$ an unseen target domain. We study zero-shot transfer: neither model parameters nor target-specific thresholds are selected using samples from $\mathcal{D}_t$. Each x is a piano system or page and y is structured notation. The central difficulty is that methods emit heterogeneous sequences. We therefore project every valid output to a multiset of pitch–duration events and compute corpus-level precision, recall, and F1. If M is the multiset intersection count, P and G the predicted and reference event counts, then

$$\text{EventF1} = 100 \frac{2M}{P + G}. \quad (3)$$

This projection measures content recovery without rewarding a method for its native tokenization. It deliberately ignores event order, voice, and hierarchy; we retain SER and normalized tree-edit distance (TEDn) on OLiMPiC to measure those stricter properties.

The protocol has four safeguards. First, all methods receive the same target images and deterministic preprocessing.

Second, no target test image is used for training, threshold selection, or checkpoint selection. Third, empty and failed outputs remain in the denominator. Fourth, all baseline values in the unified tables come from local inference and conversion, rather than copied headline numbers. We use 10,000 paired bootstrap resamples over pages or systems and report 95% intervals for F1 differences. A difference is called significant only when its interval excludes zero.

4.2 Setup and Baselines

Polish test24 and Debussy transfer24 are the main zero-shot targets. Debussy contains 24 evenly spaced systems from 24 distinct source documents. RIGOR-OMR, SMT (Ríos-Vila, Calvo-Zaragoza, and Paquet 2024), and Zeus (Mayer et al. 2024) receive identical target images and no target-domain adaptation. SMT uses its released FP-GrandStaff checkpoint and Zeus its released GrandStaff-LMX checkpoint. We additionally evaluate 200 frozen GrandStaff and 200 OLiMPiC outputs for relation ablation. All Event-F1 values are corpus aggregates; confidence intervals use 10,000 paired bootstrap resamples.

Table 1: Evaluation domains and their experimental roles.

Dataset	Target domain	<i>N</i>	Role
Polish	historical scans	24	main comparison
Debussy	unseen editions	24	comparison/ablation
GrandStaff	clean rendered	200	relation transfer
OLiMPiC	mixed systems	200	strict/relations

4.3 Comparative Results: Zero-Shot Cross-Domain Content Recovery

Table 2 presents the central cross-domain comparison. On Polish, RIGOR-OMR recovers 1,589 matching pitch–duration events and achieves an Event-F1 of 17.083, outperforming SMT and Zeus by 13.352 and 15.751 points, respectively. The corresponding 95% confidence intervals are [10.891, 15.774] and [13.477, 18.422], confirming statistically significant improvements over both baselines.

On Debussy, RIGOR-OMR achieves an Event-F1 of 8.517, exceeding SMT by 6.449 points with a 95% confidence interval of [4.200, 8.585]. It also achieves performance comparable to the strongest baseline, Zeus, demonstrating stable cross-domain applicability across substantially different visual styles and engraving conditions.

Table 2: Zero-shot pitch–duration content recovery on Polish and Debussy.

Target	Method	Prec.	Rec.	F1	Pred./Gold
Polish	RIGOR-OMR	18.734	15.700	17.083	0.838
	SMT	2.733	5.879	3.731	2.151
	Zeus	3.971	0.800	1.332	0.202
Debussy	RIGOR-OMR	7.058	10.737	8.517	1.521
	SMT	1.277	5.432	2.068	4.252
	Zeus	7.463	8.126	7.781	1.089

The output-to-reference ratios further reveal distinct recognition behaviors. On Debussy, SMT produces more than

four times the reference event count, indicating substantial over-generation. On Polish, Zeus recovers only about one fifth of the reference event count, resulting in limited recall. In contrast, RIGOR-OMR produces output volumes that are closer to the reference distributions on both domains and maintains a more balanced precision–recall profile. These results indicate that staff-normalized geometry and explicit relation reasoning provide more stable music-content recovery under substantial domain shift. Figure 2 summarizes both aggregate performance and paired uncertainty, while Appendix Figure A1 provides the corresponding per-item distributions.

4.4 Strict Structured Evaluation

Order-invariant content recovery does not capture voice ordering or hierarchical structure. Table 3 therefore reports official-style evaluation on OLiMPiC test200 using stricter sequence- and tree-structure metrics. RIGOR-OMR successfully completes all 200 conversions and consistently outperforms SMT, demonstrating that its relation reasoning and reconstruction backend can produce stable, evaluable structured outputs in an unseen score domain. The results further show that RIGOR-OMR extends beyond local pitch–duration recovery by organizing detected symbols into complete score representations, while leaving additional room for improving global ordering and hierarchical reconstruction.

Table 3: Strict OLiMPiC test200 results. Lower SER/TEDn and higher Event-F1 are better.

Method	SER	No tup.	Event-F1	TEDn
RIGOR-OMR	74.421	73.574	8.786	61.342
SMT	100.000	–	0.000	100.000
Zeus	56.486	50.625	48.242	47.751

4.5 Component Ablation: Relation Reasoning Drives the Main Gain

We evaluate all $2^4 = 16$ combinations of four components on 24 Debussy systems, resulting in 384 predictions. The components are staff geometry (A), symbol detection (B), SAM2 mask refinement (C), and explicit relation reasoning (D). Table 4 reports representative configurations, while the complete factorial results are provided in the appendix.

The full A+B+C+D configuration achieves an Event-F1 of 8.517. Removing relation reasoning reduces Event-F1 to 3.938, a decrease of 4.579 points with a paired 95% confidence interval of [2.288, 7.019]. Recall simultaneously drops from 10.737 to 2.568, showing that detecting isolated notation symbols is insufficient for recovering complete musical events. By connecting noteheads with stems and beams, explicit relation reasoning binds pitch and duration evidence to the same event and constitutes the primary source of the full framework’s improvement.

By contrast, removing SAM2 mask refinement changes Event-F1 only from 8.517 to 8.544, a difference of +0.027 points with a confidence interval of [−0.614, 0.793]. This indicates that SAM2 primarily improves the granularity and inspectability of intermediate geometric evidence, whereas

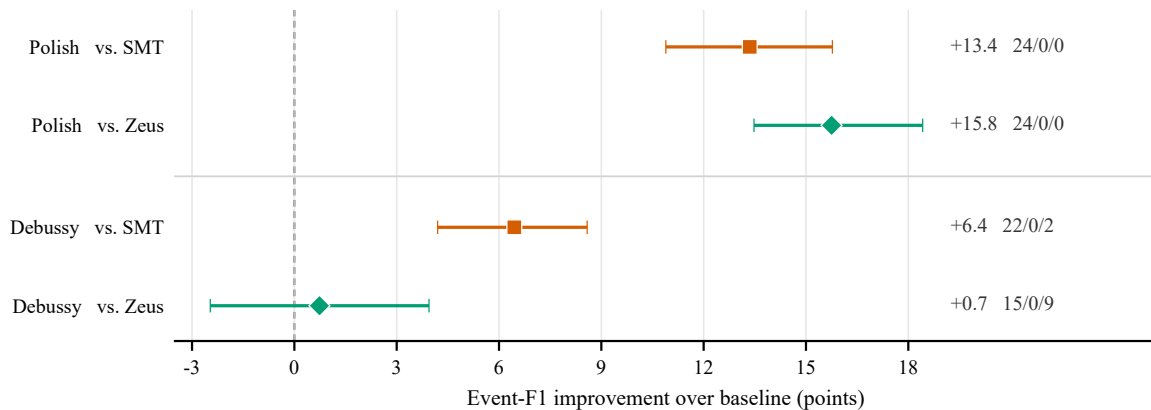


Figure 2: Paired Event-F1 improvements of RIGOR-OMR over each baseline. Points show corpus differences and bars show paired 95% bootstrap intervals; right annotations report the mean difference and per-item wins/ties/losses.

the observed content-recovery gain is driven mainly by explicit relation reasoning.

Table 4: Debussy full-factorial ablation. A: geometry; B: detection; C: SAM2; D: relations.

Variant	Precision	Recall	Event-F1
A+B+C+D	7.058	10.737	8.517
A+B+D (without C)	7.113	10.695	8.544
A+B+C (without D)	8.437	2.568	3.938
A+B (without C,D)	8.437	2.568	3.938
Without A or B	0.000	0.000	0.000

4.6 Which Relations Transfer Across Domains?

To further isolate the role of relation reasoning, we remove one relation family at a time from the frozen full-pipeline outputs and deterministically reconstruct the score without rerunning the visual modules. Compared with disabling the entire relation graph, the edge-level ablations in Table 5 identify the contribution of individual relation types more precisely.

The notehead–stem relation consistently improves performance across all three datasets. Removing it decreases Event-F1 by 1.470 points on Debussy, 0.669 points on GrandStaff, and 1.911 points on OLiMPiC.

The beam–stem relation also provides clear gains on Debussy and OLiMPiC, where its removal decreases Event-F1 by 1.403 and 1.543 points, respectively. On GrandStaff, removing this relation likewise reduces Event-F1 by 0.583 points, further indicating that beam–stem associations support cross-domain duration recovery.

Removing slur/tie endpoint relations does not change Event-F1 on Debussy because the current event metric evaluates pitch–duration content and does not include higher-level slur or tie structure.

Table 5: Relation effects under frozen visual predictions. Δ is full minus ablated Event-F1; bold intervals exclude zero. Full scores appear in Appendix Table 9.

Target	Removed edge	Δ	95% CI
Debussy	notehead–stem	1.470	[0.282, 2.647]
Debussy	beam–stem	1.403	[0.212, 2.607]
Debussy	slur/tie	0.000	[0.000, 0.000]
GrandStaff	notehead–stem	0.669	[0.013, 1.321]
GrandStaff	beam–stem	0.583	[−0.067, 1.246]
OLiMPiC	notehead–stem	1.911	[1.472, 2.358]
OLiMPiC	beam–stem	1.543	[1.132, 1.972]

These ablations use the same cached A+B+C+D path within each dataset. Their OLiMPiC full value therefore belongs to the controlled ablation protocol and is not numerically pooled with the official-style value in Table 3.

4.7 Qualitative Evidence: From Staff Geometry to Typed Reconstruction

Figure 4 follows one local beam group from the highest-scoring full-pipeline Debussy system through the stored intermediate representations. The same crop is used in every panel: staff-normalized geometry localizes the notation; accepted symbols comprise two detected noteheads, two geometry-derived stems, and one detected beam; and box-prompted masks expose their foreground support. The typed graph then connects each notehead to a stem and each stem to the beam before deterministic reconstruction. This is a composed notehead–stem–beam path; the implementation does not predict a direct notehead–beam edge.

The mask and graph views are retained for auditability rather than presented as standalone accuracy evidence. In particular, removing SAM2 does not produce a verified endpoint-metric decrease under the current ablation, whereas removing relation reasoning does.

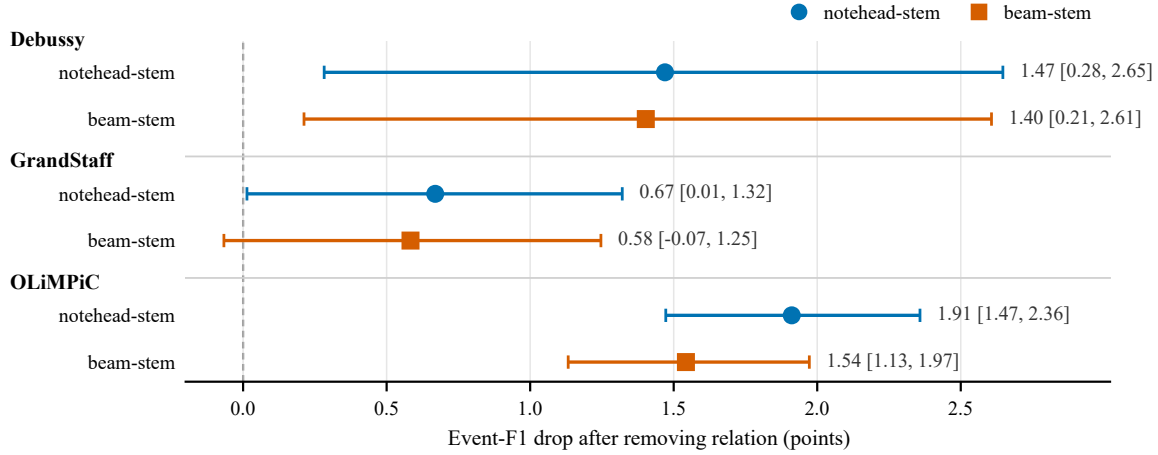


Figure 3: Event-F1 decrease after removing each relation family from frozen full-pipeline outputs. Points show corpus differences and bars show paired 95% bootstrap intervals.

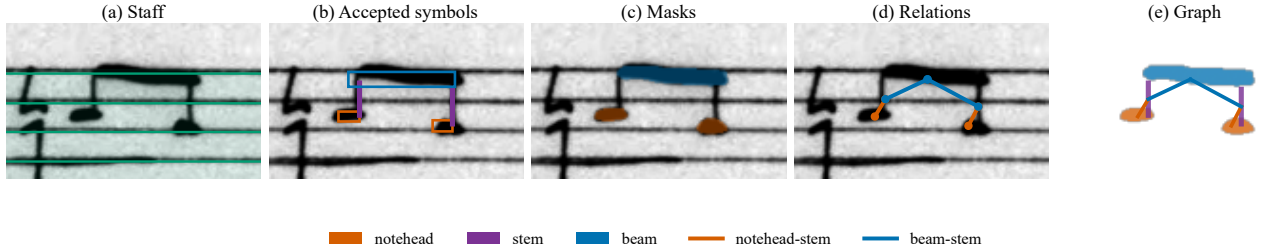


Figure 4: Qualitative evidence from the Debussy system with the highest full-pipeline Event-F1 (test_0006, 24.473). The same local beam group is followed through (a) staff localization, (b) accepted detected and geometry-derived symbols, (c) actual box-prompted SAM2 masks, (d) stored notehead-stem and beam-stem relations, and (e) graph-only reconstruction. Masks expose intermediate geometry; the ablation results do not establish an endpoint-metric gain from SAM2.

4.8 Analysis

What the Results Establish The strongest evidence comes from severe zero-shot domain shift. On Polish, RIGOR-OMR recovers 1,589 matching events compared with 81 for Zeus and raises Event-F1 from 1.332 to 17.083, with a paired confidence interval excluding zero. Debussy independently confirms a substantial improvement over SMT and performance comparable to Zeus. Edge-level ablations further attribute these gains to explicit relation reasoning: staff-normalized notehead-stem relations transfer consistently across three target domains, while beam-stem relations improve duration recovery across all three evaluated datasets. These gains require neither target-domain sequence training nor rerunning the visual modules.

Together, the results reveal complementary strengths across evaluation settings. Explicit geometry and relation reasoning preserve local musical content under substantial appearance shift, while the stricter OLiMPiC evaluation highlights further opportunities in global voice ordering and hierarchical reconstruction.

Structural Diagnosis Beyond Serialization The Polish proxy evaluation further localizes the available structural gains. Replacing the legacy staff-block serializer with a

canonical system-aware order reduces SER by only 0.1778 points, indicating that coarse serialization alone accounts for a small portion of the total error. In contrast, page-constrained diagnostics reveal substantially larger improvement spaces associated with staff assignment, within-page correspondence, and pitch-duration binding. These privileged lower bounds are used only for diagnosis and are not reported as deployable system performance. Appendix Figure A3 visualizes this diagnostic ladder and Appendix Table 8 provides the complete ledger.

Inspectability and Improvement Directions RIGOR-OMR retains staff geometry, symbol boxes, masks, typed relations, event hypotheses, and MusicXML outputs, enabling failures to be localized to the stage where information is first lost. On complex pages, low-confidence relation edges may propagate into duration or export decisions. This observation motivates confidence-aware relation pruning and globally constrained reconstruction, which can further exploit the strong local pitch and duration evidence already recovered by the framework.

4.9 Limitations and Responsible Use

Absolute zero-shot Event-F1 remains low, so the reported gains do not imply deployment-ready transcription. Event-F1 ignores order and hierarchy, and the OLiMPiC results show that these omissions are substantial. SAM2 has no verified end-metric benefit, slur/tie edges are currently inactive under the event projection, and detector replacement does not imply detector-insensitive performance. The 24-item Polish and Debussy sets also motivate broader evaluation. Finally, we have not completed a matched-hardware comparison of runtime, peak memory, or trainable parameters, so RIGOR-OMR’s modular adaptation cost remains a design property rather than an experimentally established efficiency advantage.

The datasets contain public-domain or research-licensed score images, but deployment on contemporary editions must respect copyright and dataset licenses. Inaccurate pitch or rhythm transcriptions should not be presented as authoritative; practical use should preserve the source image, expose confidence, and support human correction.

5 Conclusion

RIGOR-OMR studies OMR applicability under zero-shot domain shift rather than extrapolating from in-domain scores. Under a common pitch–duration event projection, it significantly outperforms SMT and Zeus on Polish and significantly outperforms SMT while matching Zeus on Debussy. Full-factorial and edge-level ablations attribute these gains to explicit relation reasoning, especially notehead–stem and beam–stem attachments. At the same time, OLiMPiC SER and TEDn show that local relational correctness has not yet become globally ordered, hierarchical notation. The resulting research direction is clear: preserve the transferable geometry and inspectability of RIGOR-OMR while adding constrained event binding and voice-aware structural decoding.

References

- Calvo-Zaragoza, J., Jr., J. H.; and Pacha, A. 2021. Understanding Optical Music Recognition. *ACM Computing Surveys*, 53(4): 77:1–77:35.
- Calvo-Zaragoza, J.; and Rizo, D. 2018. Camera-PrIMuS: Neural End-to-End Optical Music Recognition on Realistic Monophonic Scores. In *Proceedings of the International Conference on Music Information Retrieval*, 248–255.
- Carion, N.; Massa, F.; Synnaeve, G.; Usunier, N.; Kirillov, A.; and Zagoruyko, S. 2020. End-to-End Object Detection with Transformers. In *European Conference on Computer Vision*, 213–229.
- Ganin, Y.; Ustinova, E.; Ajakan, H.; Germain, P.; Larochelle, H.; Laviolette, F.; Marchand, M.; and Lempitsky, V. 2016. Domain-Adversarial Training of Neural Networks. *Journal of Machine Learning Research*, 17(59): 1–35.
- Huang, S.; Lu, Z.; Cun, X.; Yu, Y.; Zhou, X.; and Shen, X. 2025. DEIM: DETR with Improved Matching for Fast Convergence. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.
- Jr., J. H.; and Pecina, P. 2017. The MUSCIMA++ Dataset for Handwritten Optical Music Recognition. In *Proceedings of the International Conference on Document Analysis and Recognition*, 39–46.
- Kirillov, A.; Mintun, E.; Ravi, N.; Mao, H.; Rolland, C.; Gustafson, L.; Xiao, T.; Whitehead, S.; Berg, A. C.; Lo, W.-Y.; Dollár, P.; and Girshick, R. 2023. Segment Anything. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 4015–4026.
- Li, Y.; Liu, H.; Jin, Q.; Cai, M.; and Li, P. 2023. TrOMR: Transformer-Based Polyphonic Optical Music Recognition. In *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing*, 1–5.
- Lv, W.; Zhao, Y.; Chang, Q.; Huang, K.; Wang, G.; and Liu, Y. 2024. RT-DETRv2: Improved Baseline with Bag-of-Freebies for Real-Time Detection Transformer. *arXiv preprint arXiv:2407.17140*.
- Mayer, J.; Straka, M., Jr., J. H.; and Pecina, P. 2024. Practical End-to-End Optical Music Recognition for Pianoform Music. In *Proceedings of the International Conference on Document Analysis and Recognition*.
- Oquab, M.; Darcet, T.; Moutakanni, T.; Vo, H.; Szafraniec, M.; Khalidov, V.; Fernandez, P.; Haziza, D.; Massa, F.; El-Nouby, A.; Assran, M.; Ballas, N.; Galuba, W.; Howes, R.; Huang, P.-Y.; Li, S.-W.; Misra, I.; Rabbat, M.; Sharma, V.; Synnaeve, G.; Xu, H.; Jégou, H.; Mairal, J.; Labatut, P.; Joulin, A.; and Bojanowski, P. 2024. DINOv2: Learning Robust Visual Features without Supervision. *Transactions on Machine Learning Research*.
- Pan, S. J.; and Yang, Q. 2010. A Survey on Transfer Learning. *IEEE Transactions on Knowledge and Data Engineering*, 22(10): 1345–1359.
- Peng, Y.; Li, H.; Wu, P.; Zhang, Y.; Sun, X.; and Wu, F. 2025. D-FINE: Redefine Regression Task in DETRs as Fine-Grained Distribution Refinement. In *International Conference on Learning Representations*.
- Ravi, N.; Gabeur, V.; Hu, Y.-T.; Hu, R.; Ryali, C.; Ma, T.; Khedr, H.; Rädle, R.; Rolland, C.; Gustafson, L.; Mintun, E.; Pan, J.; Alwala, K. V.; Carion, N.; Wu, C.-Y.; Girshick, R.; Dollár, P.; and Feichtenhofer, C. 2025. SAM 2: Segment Anything in Images and Videos. In *International Conference on Learning Representations*.
- Rebelo, A.; Fujinaga, I.; Paszkiewicz, F.; Marçal, A. R. S.; Guedes, C.; and Cardoso, J. S. 2012. Optical Music Recognition: State-of-the-Art and Open Issues. *International Journal of Multimedia Information Retrieval*, 1: 173–190.
- Ríos-Vila, A.; Calvo-Zaragoza, J.; and Paquet, T. 2024. Sheet Music Transformer: End-to-End Optical Music Recognition Beyond Monophonic Transcription. In *Proceedings of the International Conference on Document Analysis and Recognition*, 20–37.
- Ríos-Vila, A.; Calvo-Zaragoza, J.; and Rizo, D. 2023. End-to-End Optical Music Recognition for Pianoform Sheet Music. *International Journal on Document Analysis and Recognition*, 26: 347–362.
- Ríos-Vila, A.; Calvo-Zaragoza, J.; Rizo, D.; and Paquet, T. 2026. End-to-End Full-Page Optical Music Recognition for Pianoform Sheet Music. *International Journal of Computer Vision*, 134(2).

- Shatri, E.; and Fazekas, G. 2023. Real World Music Object Recognition. *Transactions of the International Society for Music Information Retrieval*, 6(1).
- Shatri, E.; and Fazekas, G. 2024. Knowledge Discovery in Optical Music Recognition: Enhancing Information Retrieval with Instance Segmentation. *arXiv preprint arXiv:2408.15002*.
- Tuggener, L.; Elezi, I.; Schmidhuber, J.; Pelillo, M.; and Stadelmann, T. 2018. DeepScores: A Dataset for Segmentation, Detection and Classification of Tiny Objects. In *Proceedings of the International Conference on Pattern Recognition*, 3704–3709.
- Tuggener, L.; Satyawar, Y. P.; Pacha, A.; Schmidhuber, J.; and Stadelmann, T. 2021. The DeepScoresV2 Dataset and Benchmark for Music Object Detection. In *Proceedings of the International Conference on Pattern Recognition*, 9188–9195.
- Vaswani, A.; Shazeer, N.; Parmar, N.; Uszkoreit, J.; Jones, L.; Gomez, A. N.; Kaiser, L.; and Polosukhin, I. 2017. Attention Is All You Need. In *Advances in Neural Information Processing Systems*, volume 30.
- Wang, J.; Lan, C.; Liu, C.; Ouyang, Y.; and Qin, T. 2021. Generalizing to Unseen Domains: A Survey on Domain Generalization. In *Proceedings of the International Joint Conference on Artificial Intelligence*, 4627–4635.
- Yosinski, J.; Clune, J.; Bengio, Y.; and Lipson, H. 2014. How Transferable Are Features in Deep Neural Networks? In *Advances in Neural Information Processing Systems*, volume 27.

A Complete Component Ablation

Table 6 reports every configuration in the Debussy 2⁴ factorial experiment. Configurations without geometry (A) or detection (B) produce no events. Among viable variants, relation reasoning (D) changes recall and output coverage substantially, whereas SAM2 mask refinement (C) has no measurable end-metric benefit under this protocol.

Table 6: Complete Debussy full-factorial ablation. Pred./Gold is the predicted-to-reference event-count ratio.

Variant	Prec.	Rec.	F1	Pred./Gold
A0B0C0D0	0.000	0.000	0.000	0.000
A0B0C0D1	0.000	0.000	0.000	0.000
A0B0C1D0	0.000	0.000	0.000	0.000
A0B0C1D1	0.000	0.000	0.000	0.000
A0B1C0D0	0.000	0.000	0.000	0.000
A0B1C0D1	0.000	0.000	0.000	0.000
A0B1C1D0	0.000	0.000	0.000	0.000
A0B1C1D1	0.000	0.000	0.000	0.000
A1B0C0D0	0.000	0.000	0.000	0.000
A1B0C0D1	0.000	0.000	0.000	0.000
A1B0C1D0	0.000	0.000	0.000	0.000
A1B0C1D1	0.000	0.000	0.000	0.000
A1B1C0D0	8.437	2.568	3.938	0.304
A1B1C0D1	7.113	10.695	8.544	1.504
A1B1C1D0	8.437	2.568	3.938	0.304
A1B1C1D1	7.058	10.737	8.517	1.521

B Additional Diagnostic Results

Figure A1 shows the paired per-page and per-system zero-shot comparisons. Figure A4 then stratifies Debussy systems by a predeclared reference-event density tertile. RIGOR-OMR is strongest on the middle-density group, while Zeus is strongest on the highest-density group; performance is therefore not a monotonic function of event count alone.

Table 7: Debussy oracle diagnostics using target annotations. These are diagnostic upper bounds, not deployable results.

Method	Joint	Pitch corr.	Dur. corr.	Count corr.
RIGOR-OMR	8.517	33.901	26.119	11.118
SMT	2.068	16.530	4.553	5.562
Zeus	7.781	29.550	20.318	9.234

C Pitch–Duration Attribute Recovery

On Debussy, RIGOR-OMR achieves a pitch F1 of 26.119 and a duration F1 of 33.901, outperforming Zeus at 20.318 and 29.550 and SMT at 4.553 and 16.530, respectively. These results show that RIGOR-OMR recovers both core musical attributes under substantial domain shift.

We further identify 527 events with correct pitch evidence but unmatched duration and 760 events with correct duration evidence but unmatched pitch. This pool of partially recovered events provides headroom for improving joint Event-F1 through stronger attribute binding and structured decoding.

D Polish Structural Diagnostic Ledger

Table 8 reports the complete proxy-SER diagnostic ladder. R3 is retained for auditability but has only 141 complete

matches among 5002 gold onset groups. The cross-page relaxed bag is a mathematical reference only: because it permits tokens to match across page boundaries, it is not a valid page-constrained oracle.

E Relation Inventory and Reproducibility

All edge ablations reuse frozen A+B+C+D visual outputs, remove one relation family, and deterministically rebuild the structured prediction. Table 9 gives the complete scores, and Table 10 reports the number of instantiated edges and affected pages. Debussy confidence intervals use 10,000 paired bootstrap resamples with seed 20260720; the GrandStaff and OLiMPiC reproduction gates complete all 200 examples without failed pages.

Table 9: Complete edge-level relation ablation under frozen visual predictions. Δ is full minus ablated Event-F1.

Target	Removed edge	Full	Ablated	Δ	95% CI
Deb.-24	note–stem	8.517	7.047	1.470	[0.282, 2.647]
Deb.-24	beam–stem	8.517	7.114	1.403	[0.212, 2.607]
Deb.-24	slur/tie	8.517	8.517	0.000	[0.000, 0.000]
GS-200	note–stem	8.887	8.217	0.669	[0.013, 1.321]
GS-200	beam–stem	8.887	8.304	0.583	[−0.067, 1.246]
OLi.-200	note–stem	9.624	7.713	1.911	[1.472, 2.358]
OLi.-200	beam–stem	9.624	8.082	1.543	[1.132, 1.972]

Table 10: Relation inventory in frozen full-pipeline outputs. Entries are edge count / pages with an edge.

Dataset	Note–stem	Beam–stem	Slur/tie	Ledger–note
Debussy-24	3900 / 24	585 / 23	128 / 21	inactive
GrandStaff-200	14947 / 200	2395 / 192	198 / 83	4 / 4
OLiMPiC-200	30113 / 200	333 / 111	266 / 117	8252 / 200

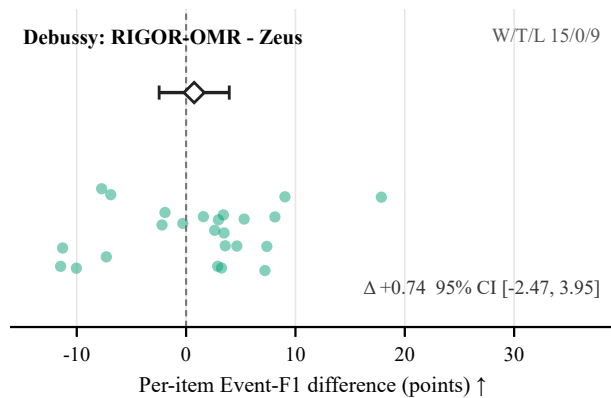
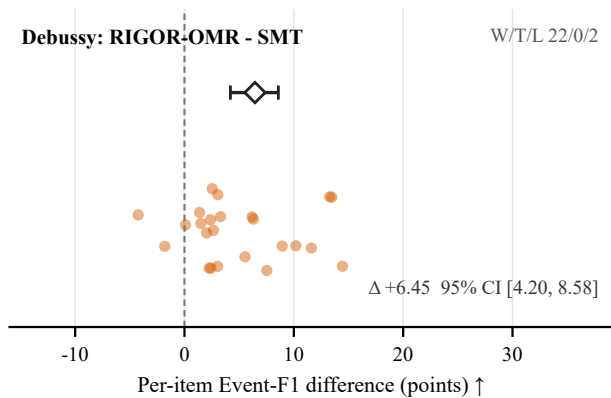
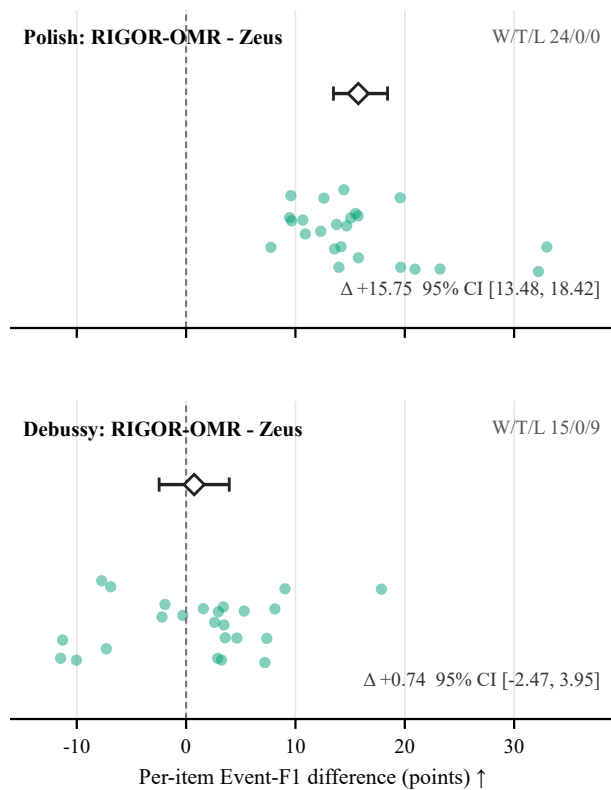
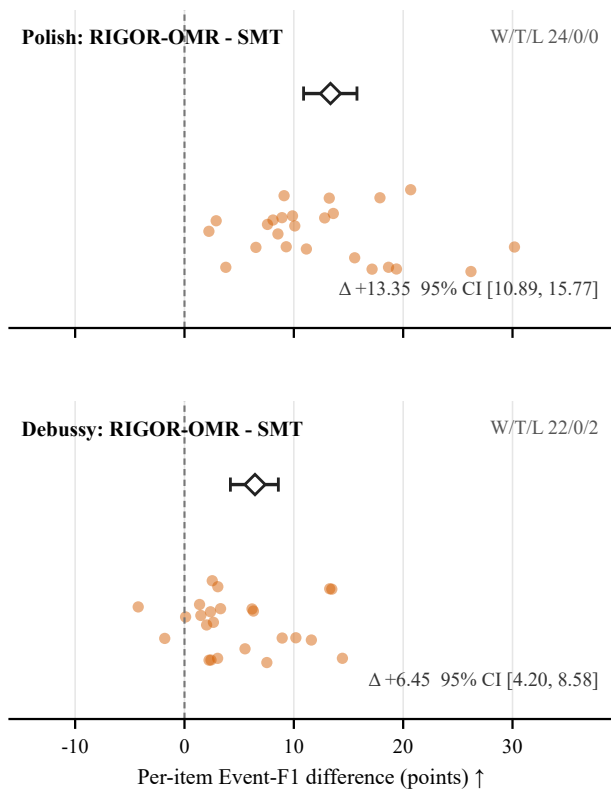


Figure A1: Per-item zero-shot Event-F1 differences. Positive values favor RIGOR-OMR. Dots are held-out pages or systems; diamonds and bars show the corpus difference and paired 95% bootstrap interval, with win/tie/loss counts reported in each panel. Polish separates cleanly from both baselines, while the Debussy comparison with Zeus overlaps zero.

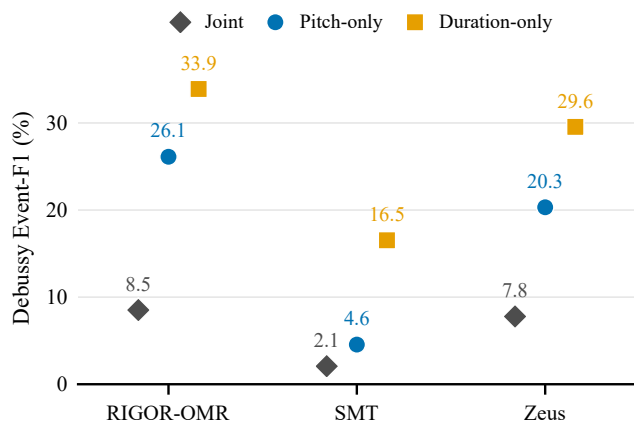


Figure A2: Joint, pitch-only, and duration-only Event-F1 on Debussy for the three frozen methods.

Table 8: Complete Polish structural diagnostics. “Gain from R0” is $SER_{R0} - SER$, so positive values denote diagnostic headroom. Lower-bound rows use privileged matching and are not deployable system outputs.

Diagnostic	Constraint	SER↓	Gain from R0	Coverage	Page-valid	Interpretation
Legacy	staff-block order	96.1367	−0.1778	corpus	yes	observed serializer
R0 canonical	system-aware order	95.9589	0.0000	corpus	yes	observed, deployable
R2 staff bag	within staff	91.7498	4.2091	corpus	yes	diagnostic lower bound
R3 exact onset	exact onset groups	95.9293	0.0296	141/5002	yes	low-coverage diagnostic
Page bag	within page	85.2584	10.7005	corpus	yes	diagnostic lower bound
R6 rebinding	within-page pitch/duration	77.1959	18.7630	corpus	yes	diagnostic lower bound
Cross-page bag	across pages	57.0793	38.8796	corpus	no	mathematical reference

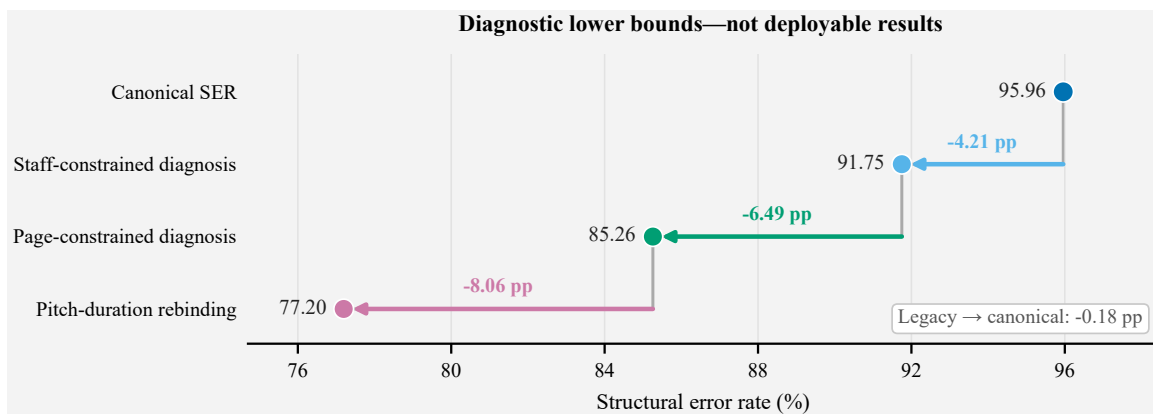


Figure A3: Polish structural diagnostic ladder. Canonical SER is deployable; subsequent values are page-constrained diagnostic lower bounds and are not deployable system results. The inset reports the legacy-to-canonical serializer change.

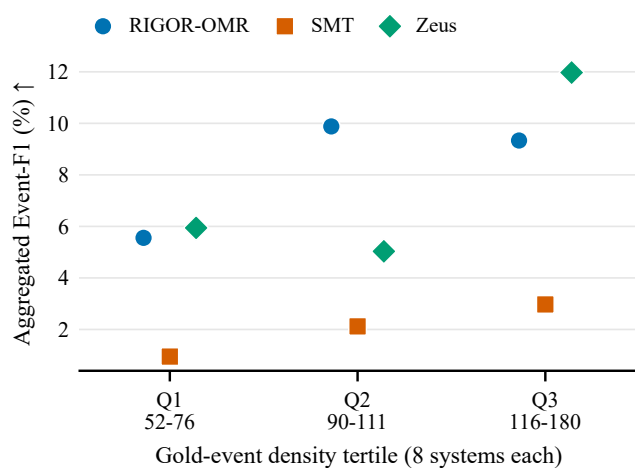


Figure A4: Exploratory Debussy density stratification. Points show aggregated Event-F1 within predeclared gold-event tertiles of eight systems each. No confidence intervals are shown because tertile-level intervals were not estimated; the non-monotonic pattern is descriptive rather than confirmatory.